

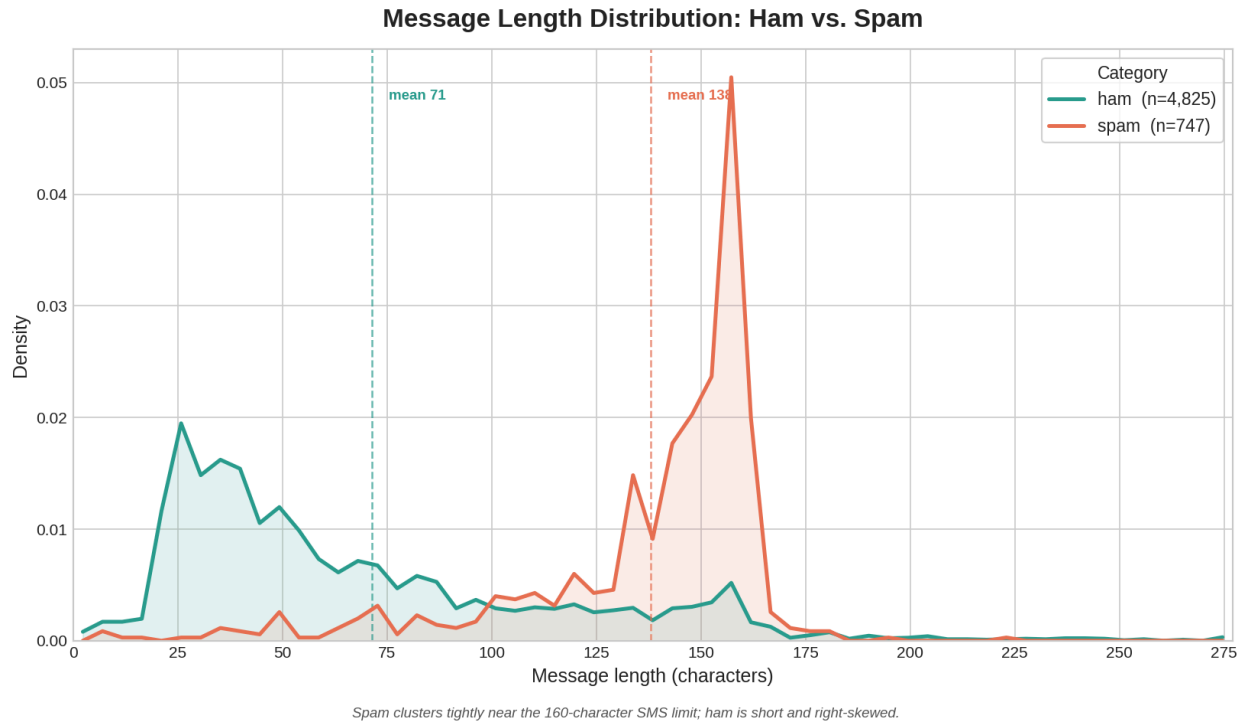
COMP 442

Group 4: Spam and phish detector

Assignment 3 - Data Visualization

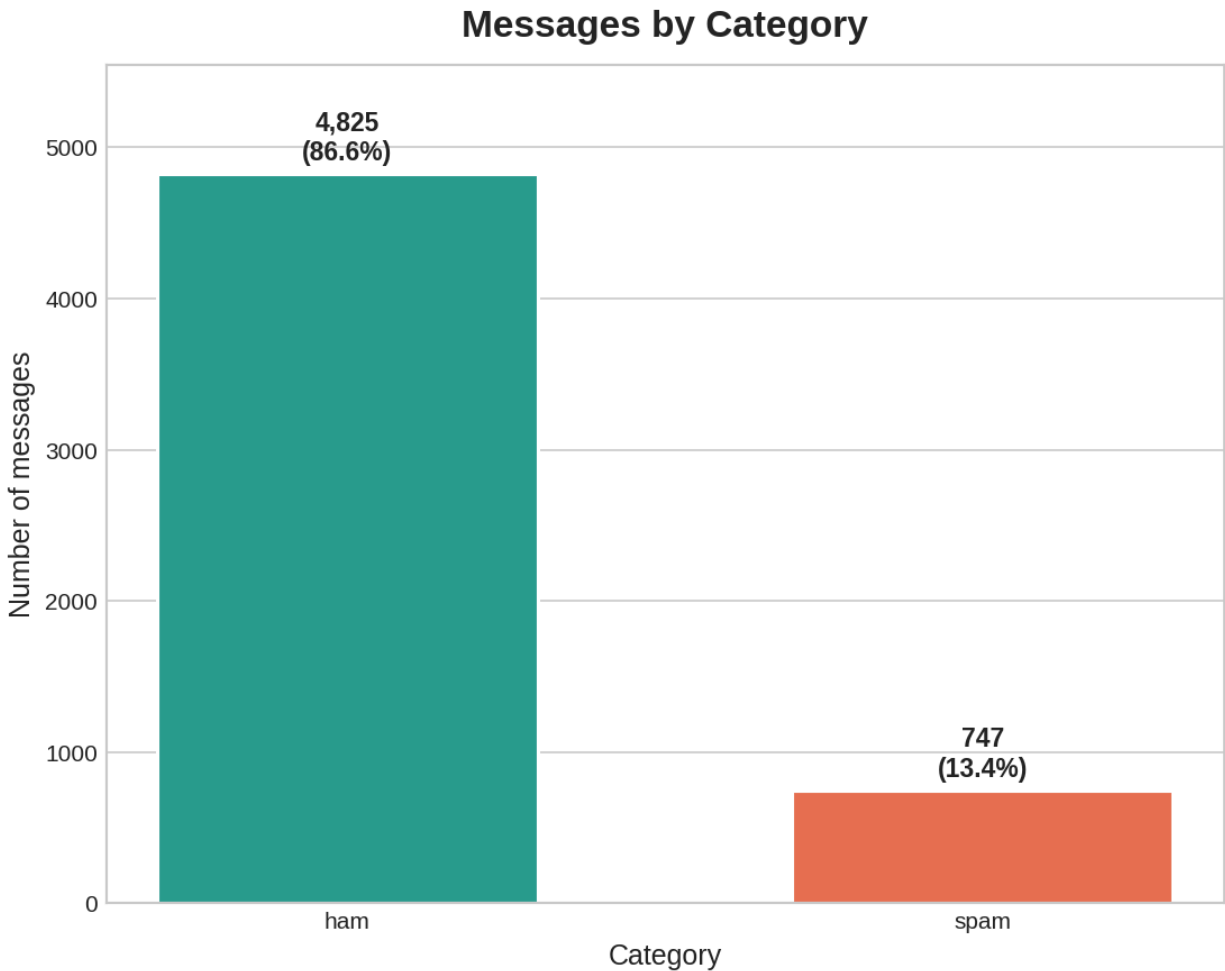
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Line plot



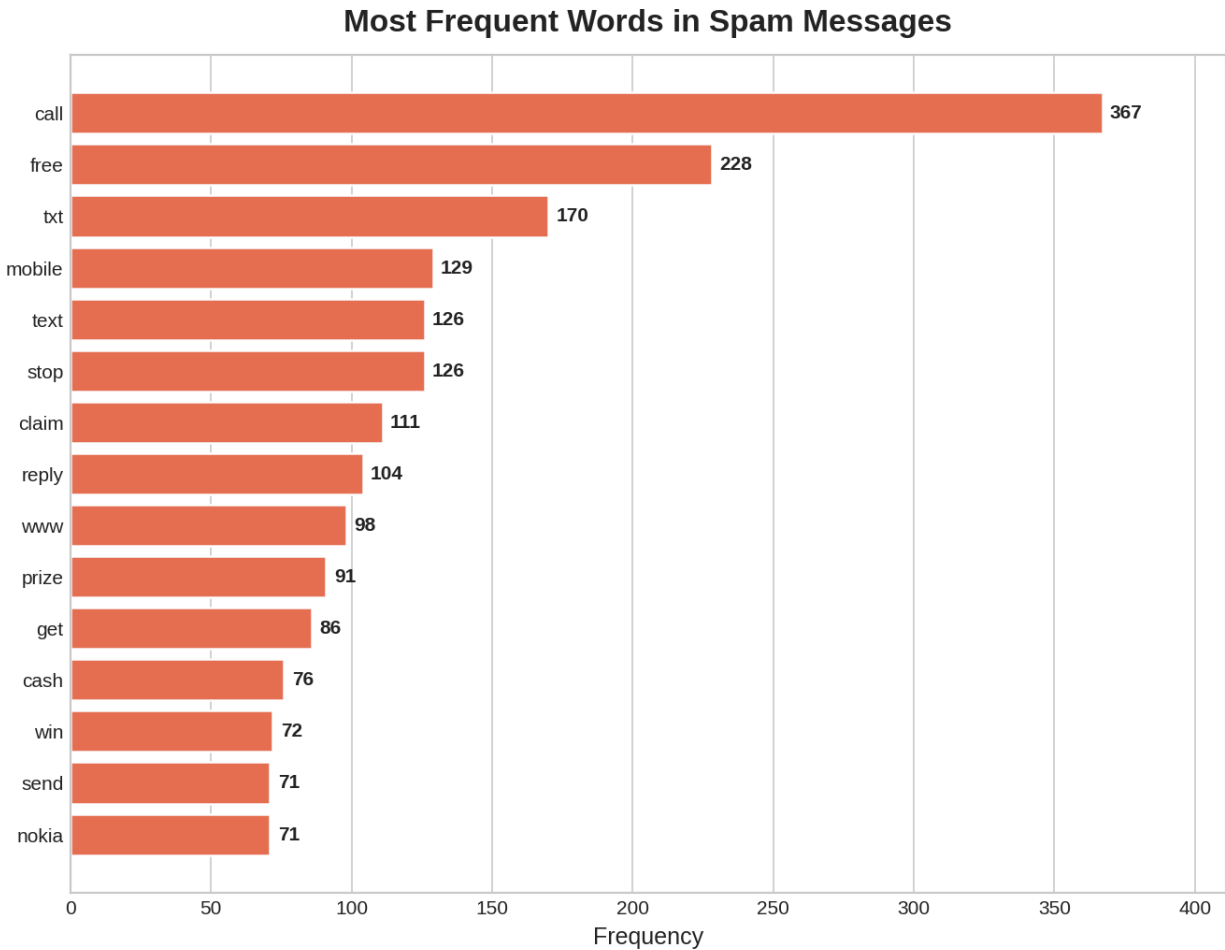
The plot indicates a clear separation and a distinct right skew to the data. Further, while it isn't a precise component (there will be others used for fine tuning the model) the spam messages peak in density showing that much larger spike clustering around 150 characters. The real, legitimate messages have a much smaller peak which can more reliably be used as a model parameter for determining legitimacy.

Vertical bar plot



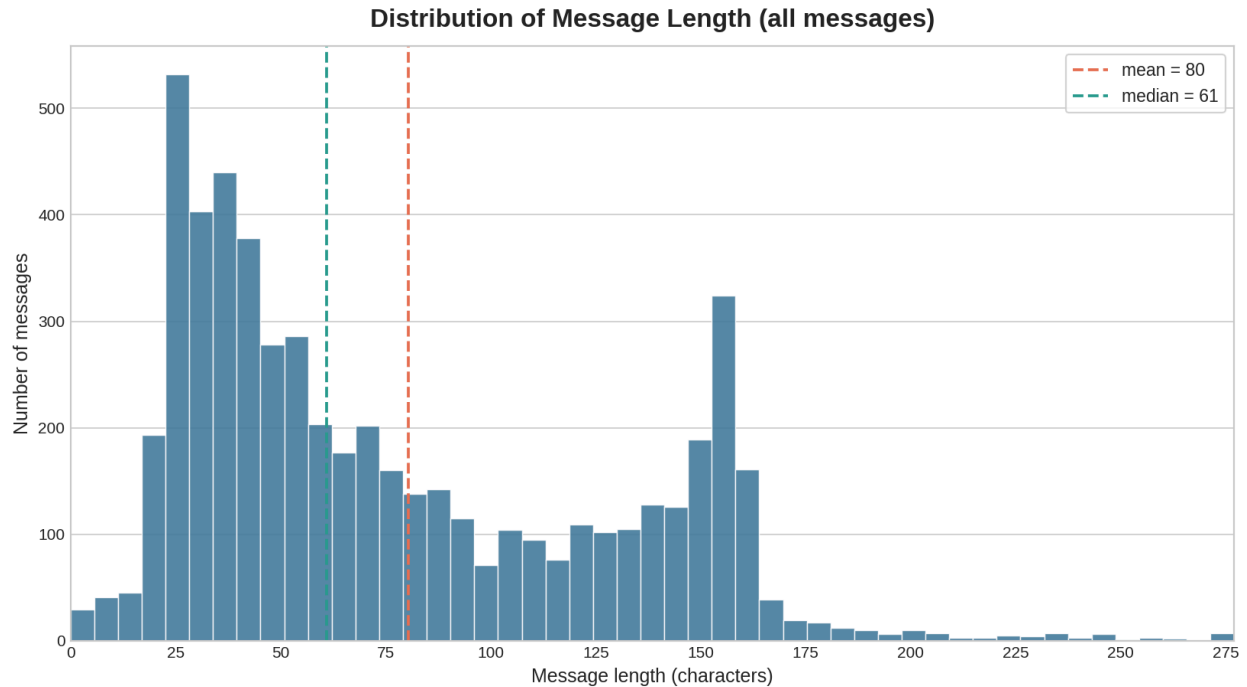
The vertical bar plot shows the distribution of messages by category. Ham messages make up the majority of the dataset, with 4825 records and spam messages account for only 747 records. This indicates a significant class imbalance in the dataset. Class imbalance is important because machine learning models may become biased toward predicting the majority class if it is not properly addressed during training.

Horizontal bar plot



This displays the most frequently occurring words in spam messages. Words such as “call”, “free”, “txt”, “claim”, and “prize” appear significantly more often than other terms, suggesting common patterns used in spam content. Many of these words are associated with promotions, rewards, or requests for user action, which are typical characteristics of spam messages. These frequent terms may serve as useful features for machine learning models when distinguishing spam from legitimate messages.

Histogram

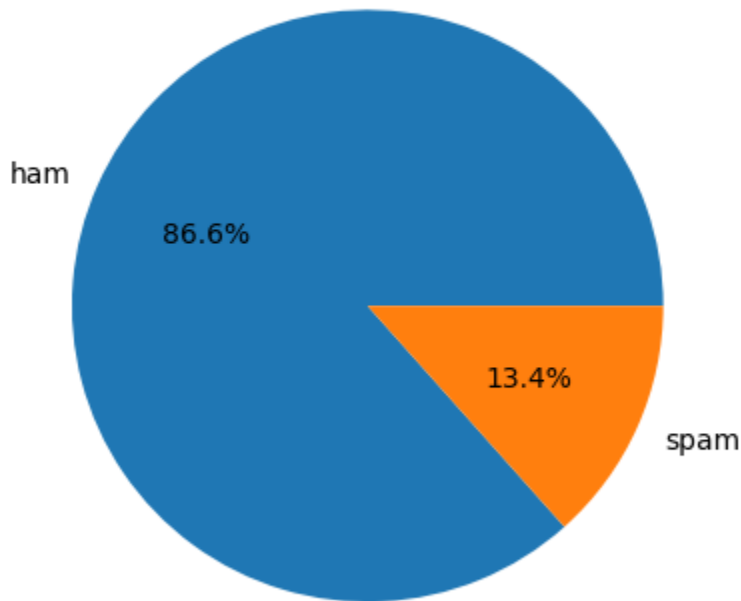


Bimodal: a tall peak of short ham messages, plus a second bump near 150 characters from spam.

This shows the distribution of message lengths across all messages in the dataset. Most messages are relatively short, with a large concentration between approximately 20 and 80 characters. The mean message length is greater than the median, indicating a positively skewed distribution. A second concentration of messages appears around 150-160 characters, while a small number of unusually long messages create a long right tail and contribute to the overall skewness of the data.

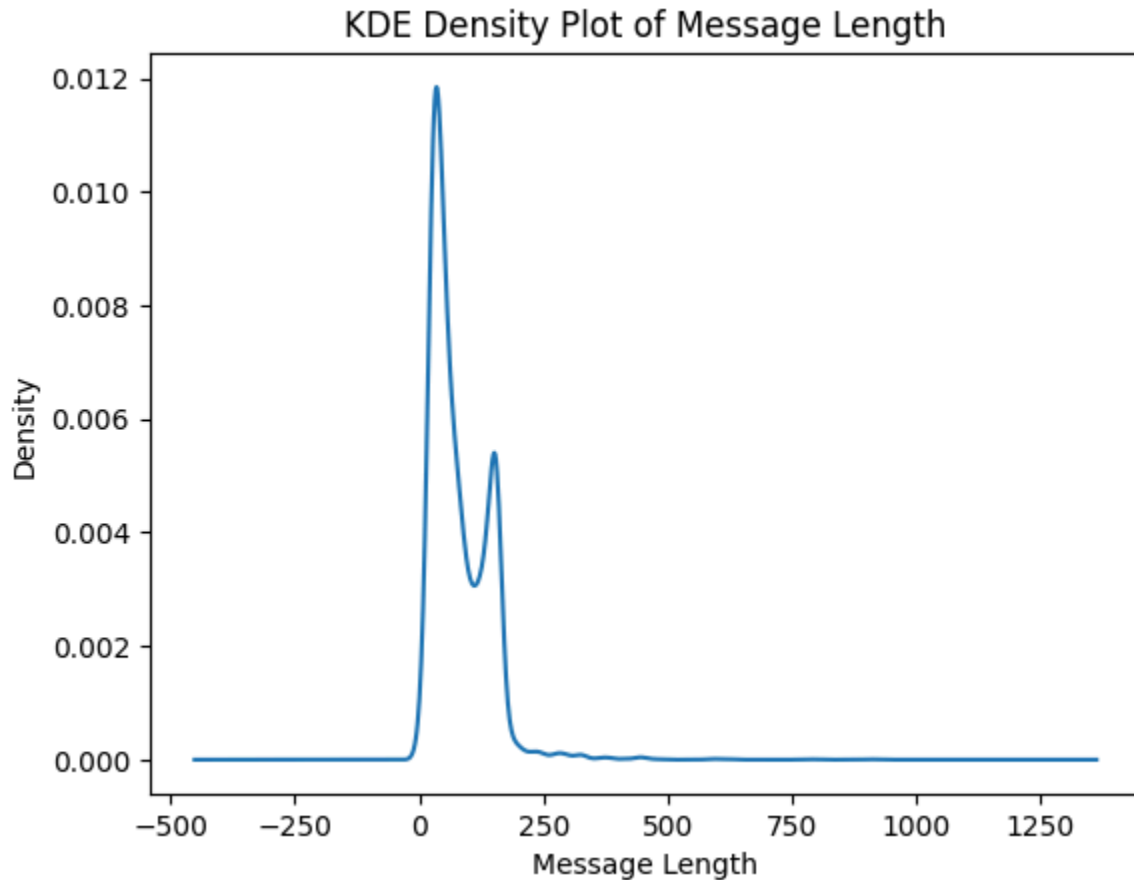
Pie plot

Spam vs Ham Percentage



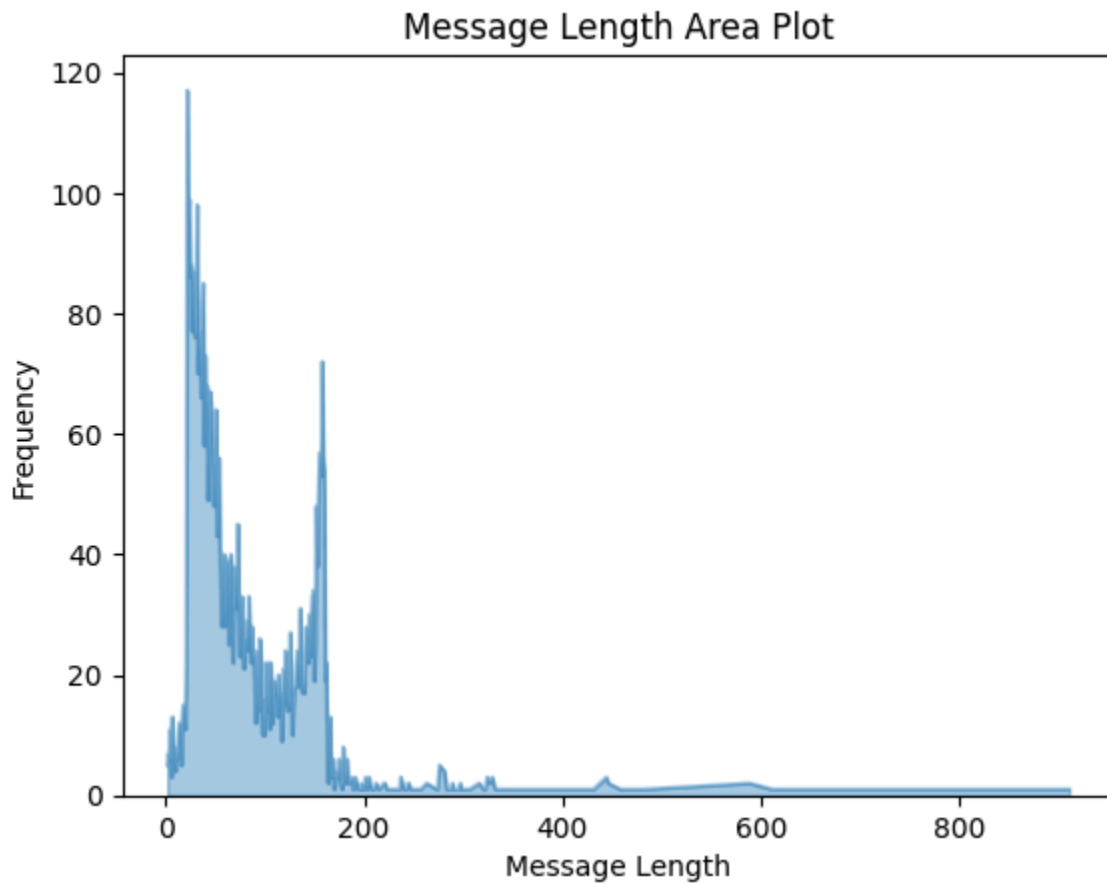
This illustrates the proportion of ham and spam messages within the dataset. Ham messages account for 86.6% of all records, while spam messages represent only 13.4%. The visualization clearly highlights the class imbalance present in the dataset, with legitimate messages significantly outnumbering spam messages. Understanding this imbalance is important because it may influence machine learning model performance and evaluation.

KDE density plot (Kernel Density Estimation plot)



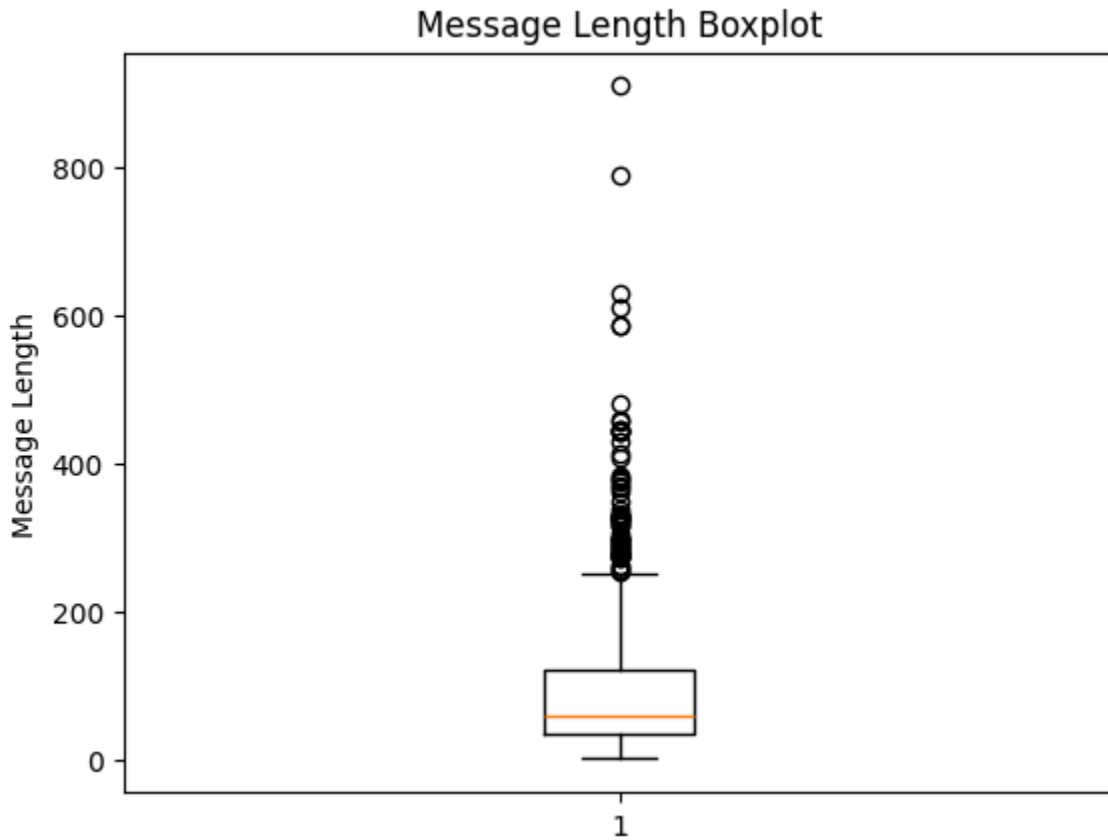
This provides a smoothed view of the message length distribution. The highest density occurs at shorter message lengths, indicating that most messages contain relatively few characters. A smaller secondary peak appears at higher message lengths, suggesting the presence of another cluster of messages. The long tail extending to the right confirms that the distribution is positively skewed and contains a small number of unusually long messages.

Area plot



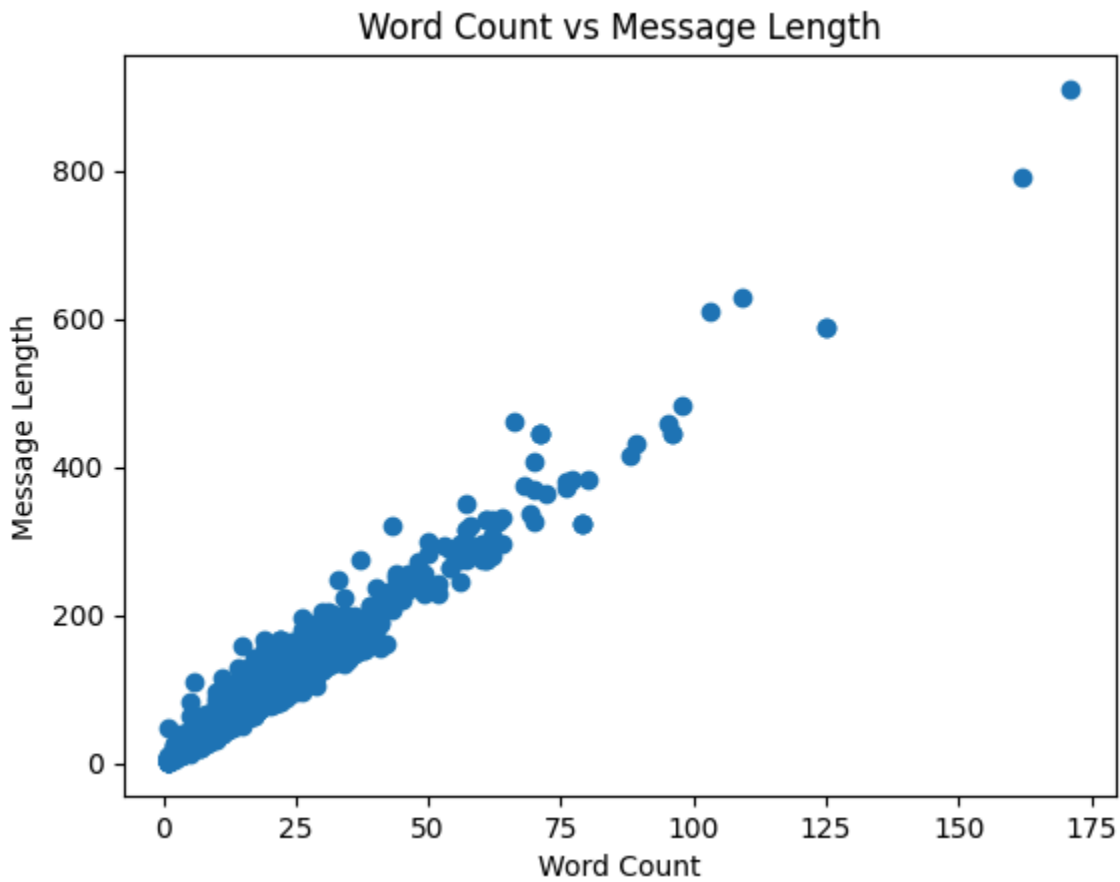
This highlights the frequency distribution of message lengths while emphasizing the overall shape of the data. Most messages are concentrated at shorter lengths, where the highest frequencies occur. The frequency decreases as message length increases, with only a small number of messages extending into much larger lengths. This visualization further supports the presence of positive skewness and confirms that long messages are relatively uncommon within the dataset.

Box (and whisker) plot



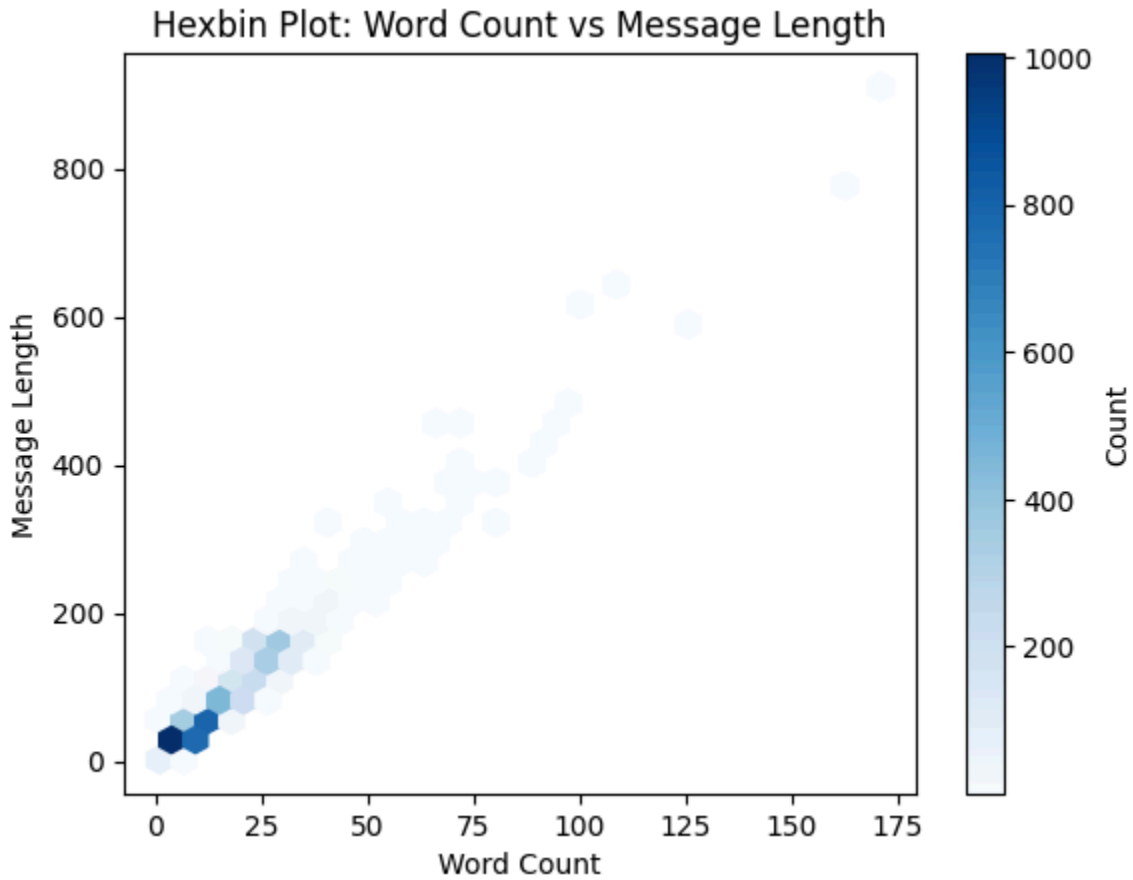
This summarizes the distribution of message lengths and highlights the presence of outliers. Most messages fall within a relatively small range, as shown by the box representing the IQR. numerous points appear above the upper whisker, indicating many unusually long messages compared to the majority of observed in previous visualizations and may influence statistical analysis and model performance.

Scatter plot



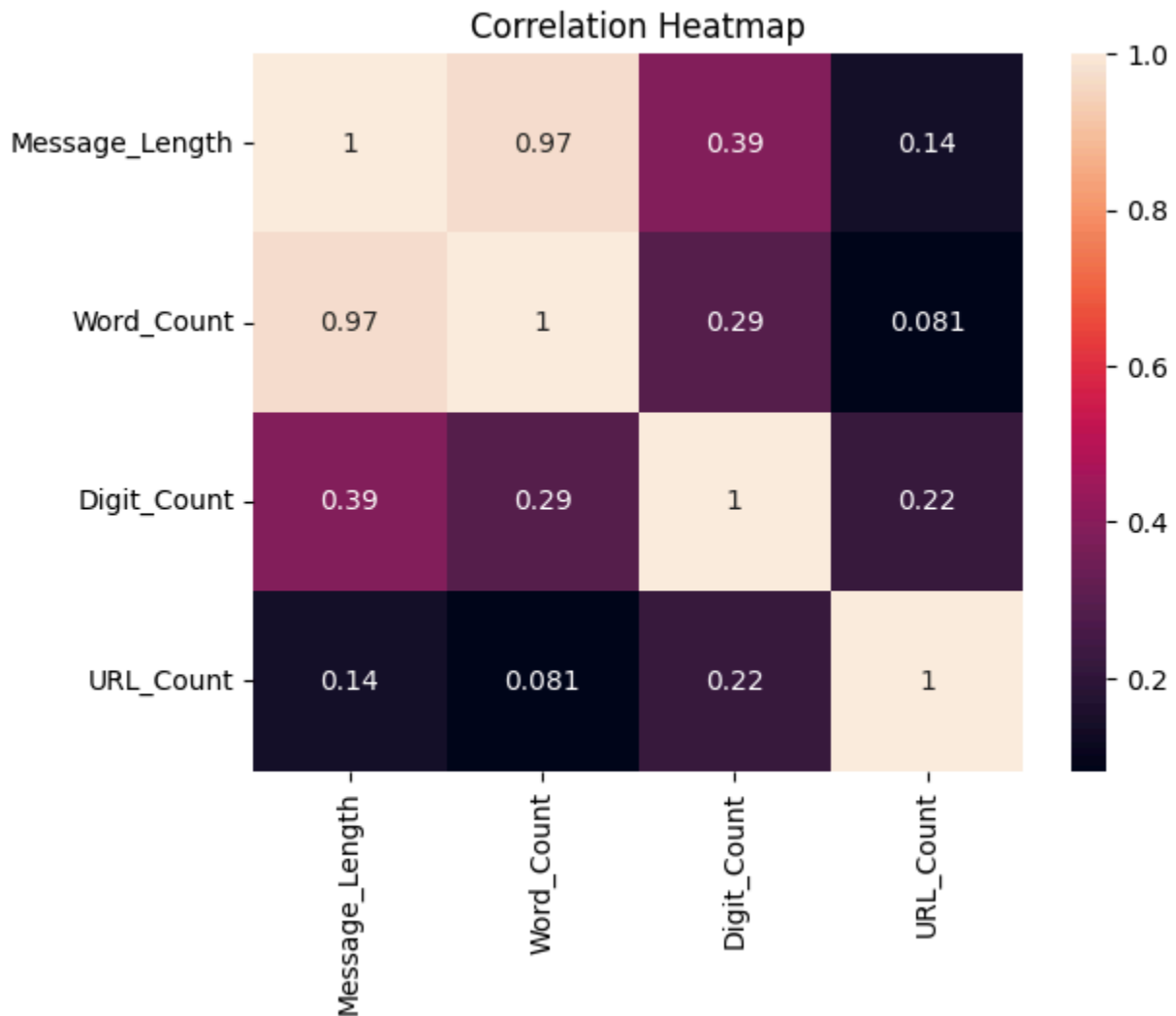
This shows the relationship between word count and message length. The points form a clear upward trend, indicating a strong positive correlation between the two variables. Messages containing more words generally have greater overall length, resulting in an almost linear relationship. A few points appear far from the main cluster, representing the unusually long messages that act as outliers within the dataset.

Hexbinplot



The hexbin plot provides a density-based view of the relationship between word count and message length. Darker hexagons represent the areas with a higher concentration of messages, while lighter regions indicate fewer observations. Most messages are clustered at lower word counts and shorter message lengths, showing where the majority of data is concentrated. The plot also confirms the strong positive relationship between the two variables and highlights a small number of outliers with unusually high word counts and message lengths.

Correlation matrix plot (heat map)



The correlation heatmap shows the strength of relationships between the numerical features extracted from the dataset. Message length and word count have a very strong positive correlation (0.97), indicating that longer messages generally contain more words. Digit count has a moderate positive relationship with message length (0.39) and word count (0.29), while URL count shows relatively weak correlations with the other features. These relationships help identify which features may be most useful for machine learning and spam classification tasks.